

Project Design Phase-II

Technology Stack (Architecture & Stack)

Technical Architecture

- **Layer 1: User Interface**
 - Data Source / Import Set Form
 - Transform Map / Field Maps Console
 - Import Log Viewer
- **Layer 2: Staging & Mapping Logic**
 - Import Set Table (sys_import_set)
 - Transform Map Engine (sys_transform_map)
 - Field Mapping Rules (sys_transform_entry)
- **Layer 3: Server-Side Processing Logic**
 - OnBefore / OnAfter Transform Scripts
 - Coalesce Field Engine
 - GlideRecord & GlideElement API
- **Layer 4: Infrastructure**
 - ServiceNow Cloud Instance
 - Personal Developer Instance (PDI)
- **Layer 5: Data Storage**
 - Target Tables (sys_user, alm_asset, incident)
 - Staging Tables (u_imp_*)
 - sys_import_set_run / sys_import_set_row
- **Layer 6: External Interfaces**
 - File Attachment / FTP / MID Server
 - REST / SOAP Integration Endpoints

Technical Documentation — Technology Stack & Application Characteristics

Table-1: Technology Stack Components

S.No	Component	Description	Technology
1	Data Staging	Import Set Table Staging	ServiceNow Import Set Framework
2	Application Logic-1	Transform Map Field Mapping	ServiceNow Transform Map Engine
3	Application Logic-2	OnBefore / OnAfter Scripts	JavaScript / GlideRecord API
4	Application Logic-3	Coalesce Matching Rules	Unique Key Matching Engine
5	Database	Target Tables (sys_user, alm_asset)	MySQL / ServiceNow Tables
6	Cloud Database	sys_transform_map, sys_transform_entry	ServiceNow Cloud DB
7	File Storage	Raw Spreadsheet File Attachment	sys_attachment / Cloud Storage
8	External API-1	Email Notifications / Logs	SMTP
9	External API-2	Automated Data Import Feed	REST / SOAP API

S.No	Component	Description	Technology
10	Machine Learning	Predictive Field Mapping	ServiceNow AI / Predictive Intelligence
11	Infrastructure	ServiceNow Cloud PDI	Web Browser / ServiceNow Platform

Table-2: Application Characteristics

Characteristic	Description
Open-Source Frameworks	Utilizes open-source libraries and frameworks for extensibility; supports community-driven integrations and modular development
Security Implementations	Role-based access control (RBAC), encrypted data transmission via HTTPS, audit logging, and input validation to ensure data protection
Scalable Architecture	Designed with modular microservices approach; supports horizontal scaling and load balancing for growing user demand
Availability	99.9% uptime target with redundant cloud infrastructure and automated failover mechanisms for continuous service